

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08

Live sessions

90 min

Per session

04 weeks

Course duration

✓ **Runs 100% on Google Colab**
Nothing to install, works even on a Chromebook

🔗 **Built for beginners**
No prior Python or machine learning required

📁 **Live code-along on Civil Engineering datasets**
Materials · Buildings & Energy · Transport · Environment · Geotechnics

📓 **Hands-on notebooks, cross-domain assignments & solutions**
Plus certificate & recordings (till 2 weeks after the course)

Course outline

01 Python for Civil Engineers
AI/ML basics, Colab, NumPy & pandas with engineering data

02 From Raw Data to ML-Ready
Explore, clean, visualize & split engineering data

03 Your First Prediction Model
Linear regression, error metrics & cross-validation

04 Trees & Ensemble Models
Decision Trees, Random Forest & boosting for nonlinear problems

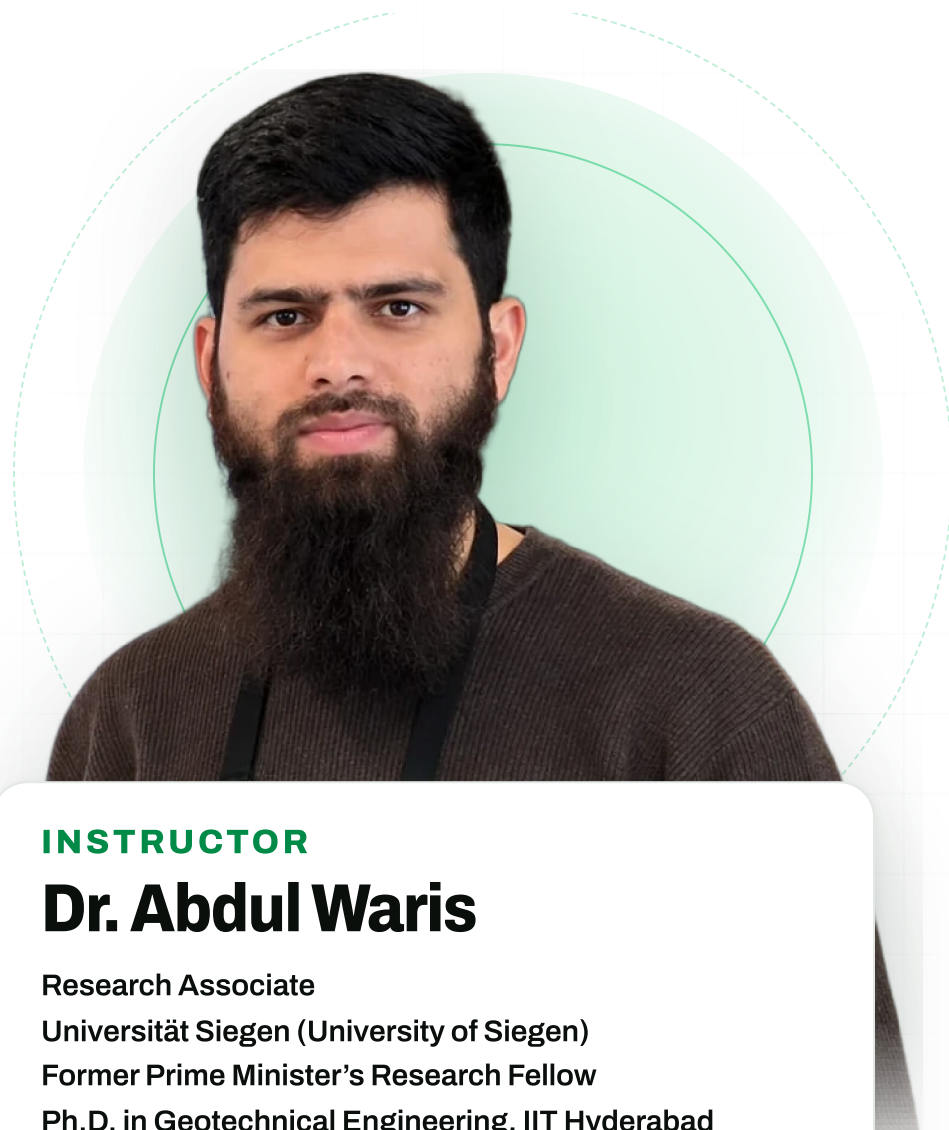
05 Classification for Engineering Decisions
Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks
Architecture, activations, training & overfitting

07 Explainable AI & Engineering Judgement
Feature importance, SHAP, and engineering interpretation

08 From Prediction to Probability
Monte Carlo, ML surrogates & probability of failure

🏗️ **Designed for Civil Engineers across disciplines** — Materials, Structures, Geotechnics, Transportation, Water & Environment.



INSTRUCTOR

Dr. Abdul Waris

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad

🏆 Research Excellence Award 2026, IIT Hyderabad

📢 Invited talks at Oregon State University (USA) and a Faculty Development Programme at D.Y. Patil (Pune, India)

🎧 Resource person, five-day ML workshop at NIT Srinagar

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty /
Practising engineers / Complete beginners to AI & ML



**Saturday,
17 October 2026**

Scan to register · qudelastudios.com →

